Gold Stock Price Forecast

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1 Introduction

Gold has long been regarded as a critical asset in financial markets. Understanding and forecasting gold stock price movements is crucial for investors, policymakers, and market analysts. Over the past decade, gold prices have experienced significant fluctuations influenced by global economic events, including financial crises and geopolitical disruptions.

This paper focuses on analyzing and forecasting the monthly closing prices of gold stock from a dataset covering the period from January 2014 to January 2024. The research objectives of this paper is to examine how accurate are Statistical Time Series Models in forecasting in such data like gold stock prices, given their perceived stability and underlying economic complexity. For this purpose, the methods that have been used are two distinct time series forecasting classes, Autoregressive Integrated Moving Average (ARIMA), Error Trend Seasonality (ETS), and a Naive benchmark. The choice of these models is motivated to capture different characteristics such as trend, seasonality (if existing), and stochastic behavior in empirical forecasting literature. ETS outperformed ARIMA and Naive metric-wise, as it captures small variations better. However, it is important to note that all models produced relatively flat forecasts except ETS which follows a straight upward forecast.

2 Methodology

2.1 Dataset Description

The dataset utilized in this paper *"Gold stock Prices"* retrieved from Kaggle platform, which is originally compiled by Wagh, 2024 from data that are provided by Nasdaq and include key financial metrics for each trading day. It consists of seven columns with 2511 unique daily gold price observations from January 19, 2014 to January 22, 2024. For the purpose of long-term time series analysis, only `Date` and `Close` (closing price) columns were retained, as they are most relevant for capturing temporal price trends.

The daily data was aggregated into monthly frequency (renamed to `Month`) by selecting the last available price of each month (Table 1), resulting in 121 monthly observations, a commonly used approach in financial time series analysis which aligns to monthly reporting, investment, and decision-making cycles.

#	Month	Close
0	2014 Jan	1239.8
1	2014 Feb	1321.6
2	2014 Mar	1283.8
...	...	...
119	2023 Nov	2057.2
120	2023 Dec	2071.8
121	2024 Jan	2029.3

Table 1: Modified - Monthly Dataset

2.2 Data Exploration and Preperation

At a first glance on the first plot (Figure 1) the close price has been continuously performed a steady increase of 59,8%, in the period of 10 calendar years. This suggests an upward trend, while there is no obvious seasonality existence by observing the seasonality plot (Figure 2). Moreover, there is a rapid increase from late 2018 to middle of 2020. This is a possible indication of structural break existence which will be further analyzed below.

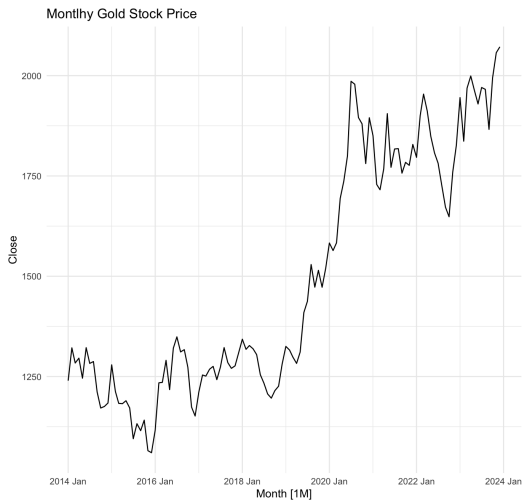

Figure 1: Plot - Raw Data

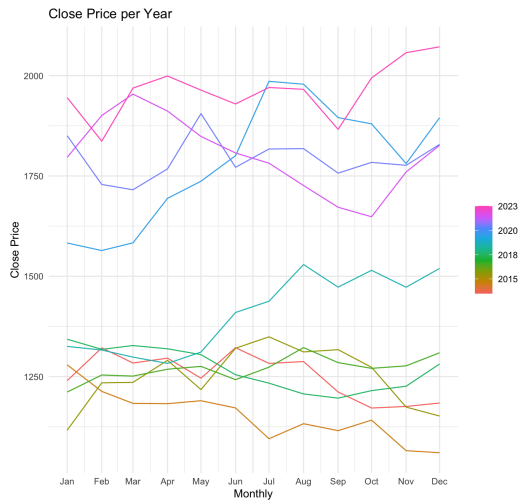

Figure 2: Seasonality Plot - Raw Data

2.2.1 Time Series Decomposition and Transformation

First and foremost, in order to analyze and modeling time series data effectively, classical time series decomposition is first applied to break down the examined time series into its primary components: trend, seasonality, and residual. Having split-up these components (Figure 3), the trend component has a steady upward trajectory over the period January 2014 – January 2024, with notable acceleration between late 2018 and mid 2020 as it has already been mentioned before. Such a surge may be related to global economic uncertainty during that period, potentially influencing gold's role as a safe asset to invest in. The seasonality component indicates a possible seasonality as it exhibits fluctuations with peaks and troughs repeating at fixed intervals throughout the series. "In classical decomposition, we assume that the seasonal component is constant from year to year" (Hyndman & Athanasopoulos, 2021), meaning that any recurring seasonal pattern has a stable magnitude and shape over time. Last but not least, the fourth panel represents white noise. Multiplicative method was also applied which showed that the graphs are almost the same.

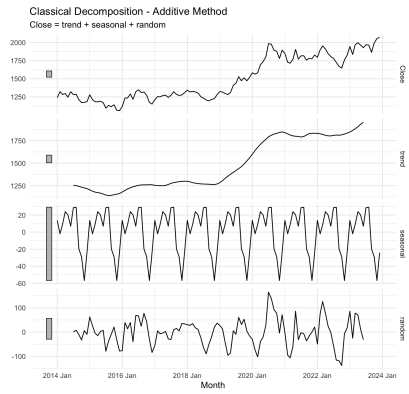

Figure 3: Classical Decomposition Additive Method - Raw data

On financial time series usually heteroskedasticity exists, where variance changes over time. To examine and address this, in order to ensure that model assumptions are met for the forecasting models later (ARIMA and ETS), first a Box-Cox transformation was applied to the raw data. Upon inspection of ACF and PACF plots of Box-Cox transformed data (Figures 4 - 5), slowly decaying autocorrelation exist, which are consistent with the presence of a strong trend component. The optimal lambda value estimated close to zero (through Guerrero). As (Nahhas, 2025) suggests, "When a transformation is needed but lambda is not far enough from 0 to make a meaningful difference, then use a log-transformation since it is commonly used and therefore understandable to a broader audience". This fact may indicates that the data are noisy and small differences in lambda will not affect results significantly. Given the near equivalence of results, log-transformed series was chosen for the rest of analysis.

Figure 4: ACF - Box-Cox transformation

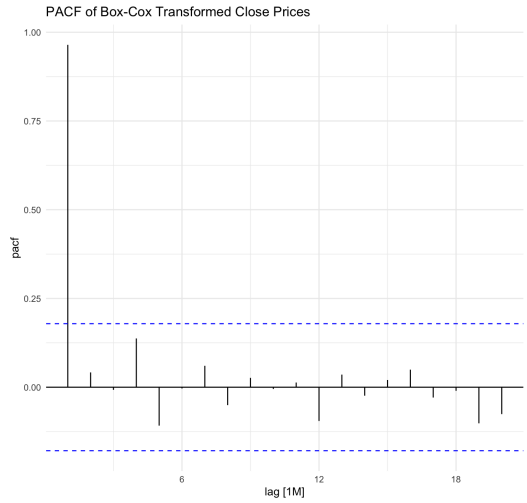

Figure 5: PACF - Box-Cox transformation

After the log transformation, STL decomposition (Seasonal and Trend decomposing using Loess) was performed (Figure 6). This choice was made due to the nature of the data, as in trading, is generally more difficult to exist a consistent or strong seasonality due to unpredictable factors, trading behavior etc. Compared to classical decomposition, STL is better in estimating trend and seasonal components, especially when seasonality is weak or variable. Again, the second panel in the log-transformed series remains strong and upward, while the seasonal component on the third panel still confirms earlier findings, that seasonality is negligible, showcasing that gold's monthly closing prices are primarily driven by long-term market forces rather than cycles.

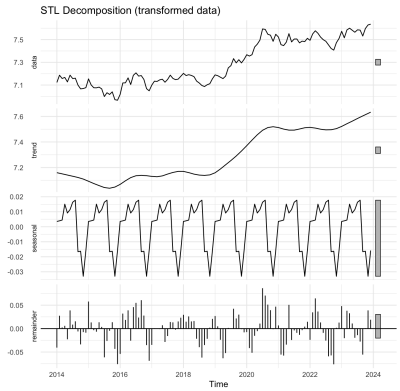

Figure 6: STL Decomposition - Transformed data

2.2.2 Stationarity

A fundamental assumption for many time series modeling techniques is stationarity, which signifies that the statistical properties of time series (such as mean, variance, etc.) remain

constant over time. As Hyndman and Athanasopoulos, 2021 noted, "Thus, time series with trends, or with seasonality, are not stationary", a crucial step in time series analysis is to assess stationarity and, if the series is non-stationary, to transform it accordingly. According to a recommendation from Tannyasharma, 2023 from Medium, "Neither the ADF test nor the KPSS test will confirm or disconfirm stationarity in isolation. Run both tests to decide if you should difference. If at least one of the tests claims to have found non-stationarity, you should difference.", both KPSS and ADF tests were performed (Table 2 & 3).

#	kpss_stat	kpss_value
1	2.19	0.01

Table 2: KPSS Test

#	Dickey-Fuller	Lag order	p-value
1	-2.7377	4	0.2702

Table 3: Augmented Dickey-Fuller Test

Both tests resulted that the series are non-stationary (KPSS p-value < 0.05 and ADF p-value > 0.05). However, KPSS is most accurate as it assumes stationarity by default and looks for evidence of non-stationarity. In other words, it tries to reject it. So, to succeed stationarity, one-time differencing was applied where trend was removed and mean was stabilized. On Figures 8 & 9, there is not a specific pattern of lags for showing correlation, instead, random spikes appear, which is sign of stationarity's existence.

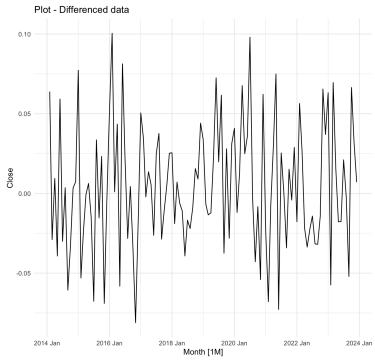

Figure 7: Transformed Autoplot

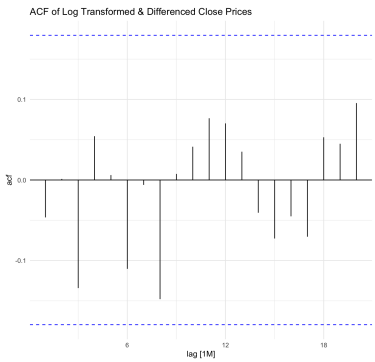

Figure 8: ACF - Log-Differenced Transformation

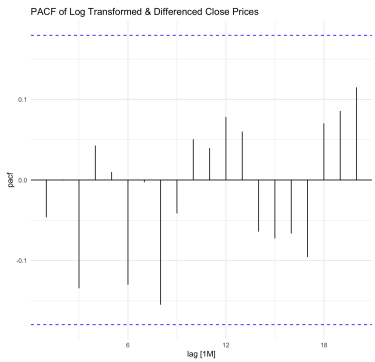

Figure 9: PACF - Log-Differenced Transformation

2.2.3 Structural breaks

Structural break is a sudden and significant change in time series. If they are not spotted effectively, they can lead to huge forecasting errors and unreliability of the model in general. There are several methods to spot and analyze structural breaks existence. "Chow test is one of the most commonly used in time series analysis to test for the presence of a structural break at a period which can be assumed to be known a priori (for instance, a major historical event such as a war)" (Wikipedia, 2025). In this analysis, the Chow test has been used to test for structural breaks at Covid-19 break (February

2020). The series were differenced to achieve stationarity at this time period, with residual diagnostics to confirm the absence of autocorrelation.

3 Results

3.1 Model choice

For specifying the forecasting models, starting by examining ACF and PACF plots of the stationary time series (Figure 8 - 9) is essential as ARIMA models rely on autocorrelation patterns.

Both ACF and PACF plots do not show significant and repetitive spikes for seasonality indication. In ARIMA modeling, AR(p) and MA(q) term represents the number of autoregressive lags, determined by the point where PACF and ACF plots cut off respectively. Both of the stationary series' plots start with negative value at lag 1 and display no significant spikes beyond that. This lack of autocorrelation suggests that neither strong AR nor MA terms are necessary, while the absence of any seasonal patterns reinforces the choice of non-seasonal ARIMA specification. Based on these observations, p=q=0 was set with one time differencing d=1.

Interestingly, the model we manually specified happens to be the same as the aytomated model selection precedure with autoARIMA(0,1,0)(0,0,0), confirming the validity of the visual and theoretical reasoning applied. Furthermore, several alternative ARIMA models applied with parameters randomly chosen. Their performance was tested and compared using AIC, AICc and BIC. As Hyndman and Athanasopoulos, 2021 recommends, best performed ARIMA model is the one with the lowest AICc. Taking this into consideration, autoARIMA(0,1,0)(0,0,0) model seems to have the most negative AICc, and therefore be the preferable among all candidate models (Table 4).

#	**.model**	**AIC**	**AICc**	**BIC**
1	autoARIMA(0,1,0)(0,0,0)	-389	-389	-387
2	ARIMA(1,1,1)(0,0,0)	-386	-385	-378
3	ARIMA(2,1,1)(0,0,0)	-384	-384	-374
4	ARIMA(1,1,2)(0,0,0)	-384	-384	-373
5	ARIMA(2,1,2)(0,0,0)	-383	-382	-369

Table 4: ARIMA Models - Comparison

Inspecting the Table 5, p-value of Ljung Box Test is greater than 0.05 which means that there is no significant autocorrelation left in the residuals, so the model fits the data well in terms of residual independence (Figure 10).

#	.model	lb_stat	lb_pvalue
1	autoARIMA(0,1,0)(0,0,0)	6.41	0.780

Table 5: AutoARIMA - Ljung Box test

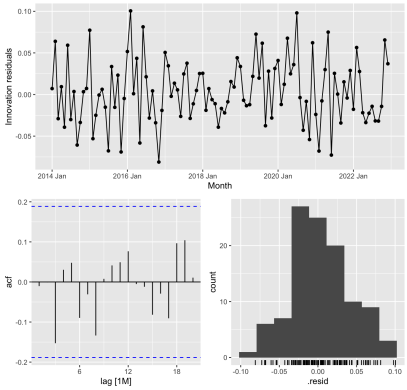

Figure 10: Residuals of auto ARIMA

Continuing to the Exponential Smoothing (ETS) models, the process began with decomposing the series to visually inspect trend and seasonality patterns as noted above on the analysis, on section "Time Series Decomposition and Transformation". In ETS modeling, the error(E), trend(T) and seasonality(S) components can be specified as additive(A), multiplicative(M) or none(N), based on the data structure. An autoETS selection procedure was applied which automatically selected ETS(A,N,N) parameters. To ensure that this choice was consistent with the data characteristics, two more manually fitted ETS models were applied with different terms. Model comparison was conducted using again the AICc criteria, where autoETS(A,N,N) model showed the lowest (most negative) AICc value confirming its suitability as the preferred forecasting model (Table 6)

#	.model	AIC	AICc	BIC
1	autoETS(A,N,N)	-191	-191	-183
2	ETS(M,M,N)	-188	-187	-174
3	ETS(A,A,N)	-188	-187	-174

Table 6: ETS Models - Comparison

3.2 Forecasts

The forecasting results are shown on Figures 11 - 13, where the predictions where made for 1 year. AutoARIMA and autoETS models illustrate a straight flat line while the manual ETS models forecasted an increasing straight line. This is a result from defined the trend component as additive and multiplicative.

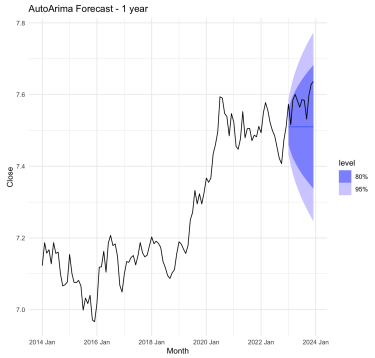

Figure 11: AutoARIMA Forecast

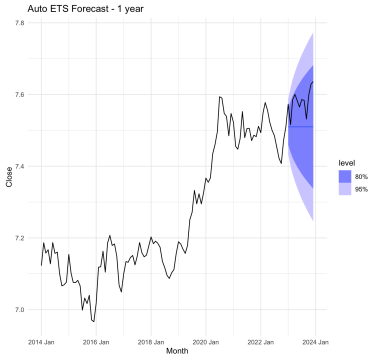

Figure 12: AutoETS Forecast

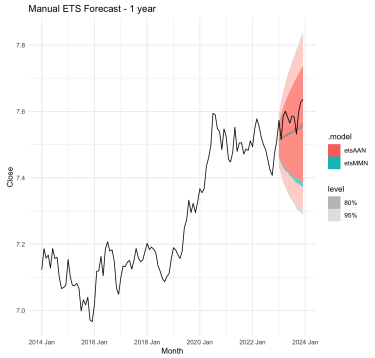

Figure 13: Manual ETS Models Forecasts

As noted on section 2.2.3, after confirming the presence of structural break on COVID-19 break using Chow test, prompting segmentation of the series into pre-break and post-break periods for separate modeling was applied. Despite fitting AutoARIMA models on each stationary segment, the resulting forecasts (Figure 14) remained essentially flat for both periods.

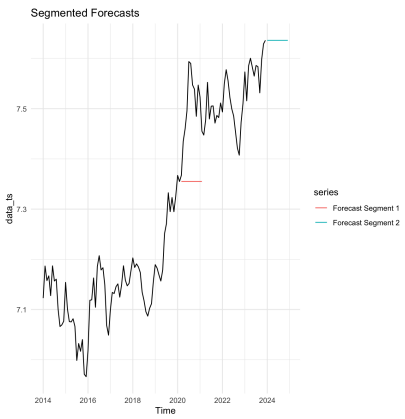

Figure 14: AutoARIMA Segmented Forecast

Evaluating the models, four significant accuracy metrics were considered: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Percentage Error (MPE), and Mean Absolute Percentage Error (MAPE). RMSE and MAE measure the average magnitude of forecast errors, with RMSE penalizing larger errors more heavily, making it more significant as measure when outliers exist, because it's more sensitive to them. MPE and MAPE express forecast errors as percentages, with MPE retaining the direction of bias (positive or negative) and MAPE using absolute values to focus solely on the magnitude of the error. Across all these metrics, lower values indicate higher forecast accuracy.

The summarized accuracy metrics from the automated models displayed on the Table 7, are very similar and higher than the rest. On the other hand, ETS(A,A,N) achieved

the lowest scores in all four metrics, indicating superior performance in capturing the underlying trend in gold prices compared to the other models. Furthermore, the additive error term and the additive trend component without seasonality, allowed it to effectively track the upward movement in gold prices and that's why it follows the existing growth pattern. Lastly, the naïve model was tested as a simple benchmark, producing a flat forecast line as well. However, its performance metrics were notably weaker, confirming its limited ability to capture the underlying trend in gold prices.

#	**.model**	**RMSE**	**MAE**	**MPE**	**MAPE**
1	autoARIMA(0,1,0)(0,0,0)	0.0794	0.0723	0.951	0.951
2	autoETS(A,N,N)	0.0796	0.0724	0.954	0.954
3	ETS(M,M,N)	0.0573	0.0515	0.658	0.678
4	ETS(A,A,N)	0.0519	0.0473	0.576	0.622
5	NAIVE	0.0794	0.0723	0.951	0.951

Table 7: Models' Performance Metrics

3.3 Discussion

Gold stock prices are heavily influenced by unpredictable external factors, resulting in high volatility and randomness that challenge the detection of consistent, strong patterns in historical data. Forecasting models such as ARIMA, ETS, and Naive, which were evaluated in this study seem to be a good baseline. After evaluation, it is spotted that they struggle to capture the complex dynamics in the series. In particular, the segmented ARIMA models fitted before and after the detected structural break behaved essentially as random walks with no significant autoregressive or moving average components, resulting in flat forecasts that merely carried forward the last observed values. This indicates that, despite statistical justification for segmentation, the structural break did not translate into meaningful changes in the underlying stationary processes. Given these limitations, future research should consider more advanced statistical or hybrid approaches that integrate machine learning or deep learning techniques, such as Gradient Boosting or LSTM networks, that may better capture the behaviors of gold stock prices and improve forecast performance.

4 Conclusion

This study examined the time series characteristics of monthly gold closing prices and the forecasting of the price of gold stock over 1 year horizon, using ARIMA and ETS models. After a first inspection of the data, classical and STL decomposition methods were used, stationarity tests, structural break analysis, and performance metrics were compared to determine the most suitable model. The results confirmed a strong upward

trend negligible seasonality and identified structural break around the COVID-19 period. ARIMA and ETS were implemented and evaluated, with additive ETS(A,A,N) model demonstrating the best accuracy. However, all tested models struggled to account for the complex volatility and changes in gold stock price dynamics, highlighting limitations of traditional statistical models in such contexts. Overall, the study provides valuable insights into gold stock price behavior and serves as a foundation for improving forecasting strategies in volatile financial markets.

References

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* [https://otexts.com/fpp3/classical-decomposition.html].

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Appendix

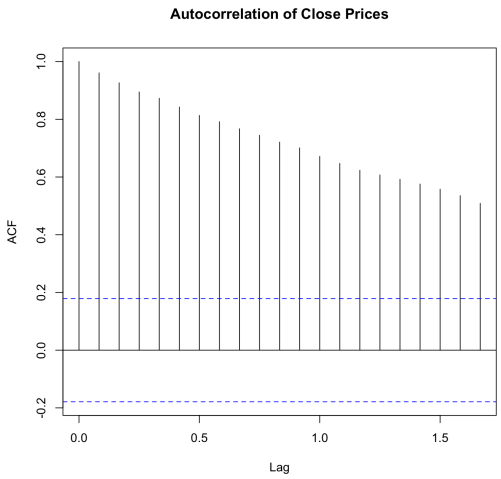

Figure 15: ACF Raw Data

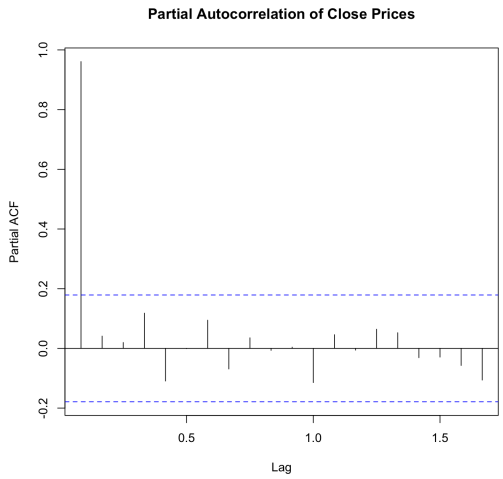

Figure 16: PACF Raw Data

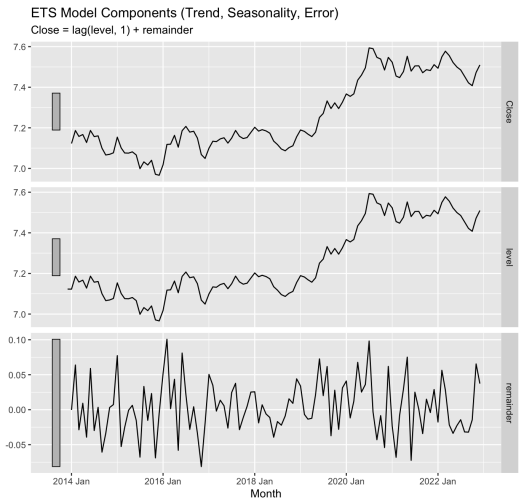

Figure 17: autoETS Components

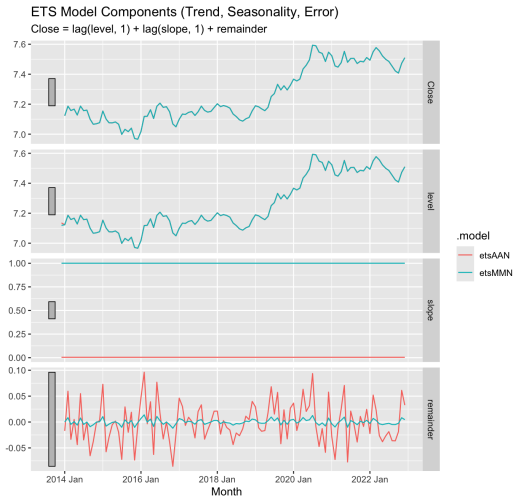

Figure 18: Manual ETS models' Components

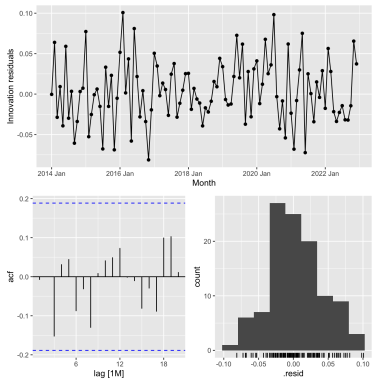

Figure 19: Residuals of Auto ETS

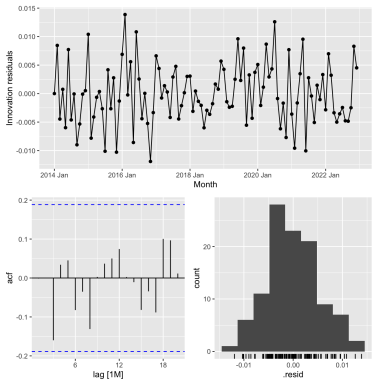

Figure 20: Residuals of ETS(M,M,N)

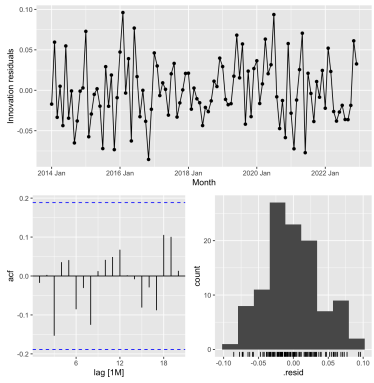

Figure 21: Residuals of ETS(M,M,N)

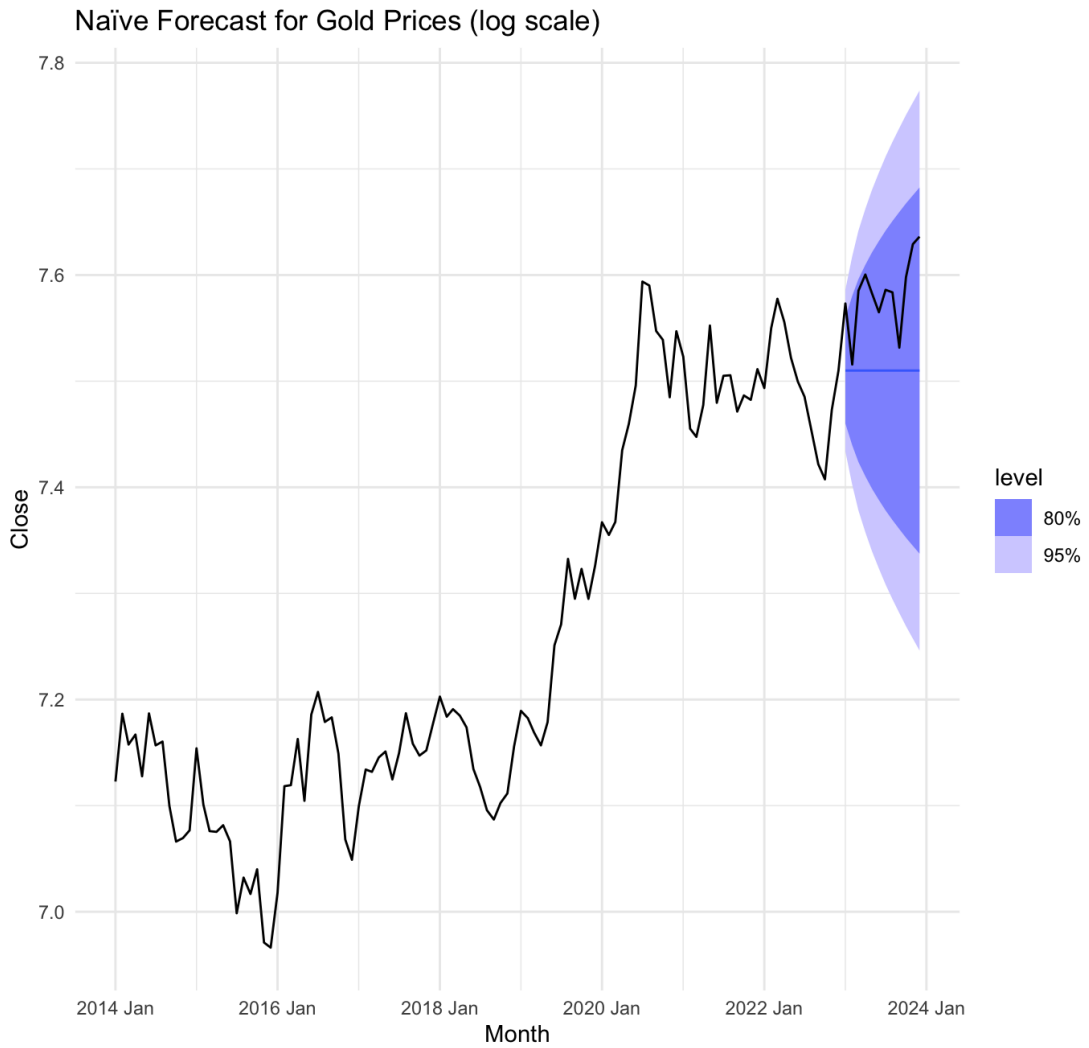

Figure 22: Naive Forecast